

LESSON 5.5

APPLYING THE TRIANGLE INEQUALITY THEOREM



LESSON INTRODUCTION

MA.8.GR.1.3 Use the Triangle Inequality Theorem to determine if a triangle can be formed from a given set of sides. Use the converse of the Pythagorean theorem to determine if a right triangle can be formed from a given set of sides.

LEARNING TARGETS

- I can use the Triangle Inequality Theorem to determine if a given set of values makes a triangle.
- I can use the Triangle Inequality Theorem to determine possible values for the third side of a triangle when given two sides.

BENCHMARK COHERENCE

BUILDING ON

N/A

WORKING TOWARD

MA.912.GR.1.3 Prove relationships and theorems about triangles. Solve mathematical and real-world problems involving postulates, relationships, and theorems of triangles.

ESSENTIAL QUESTIONS

- How can you use the Triangle Inequality Theorem to determine if a given set of values can make a triangle?
- How can you use the Triangle Inequality Theorem to determine possible values for the third side of a triangle when given two sides?

LESSON NARRATIVE

This lesson will solidify student understanding of the Triangle Inequality Theorem. In the previous lesson, students explored how the Triangle Inequality Theorem works and what it is used for. This lesson puts it into practice. Students will be given a set of three side measurements and will determine if they can form a triangle or not. Students will also be given two side measurements and determine what a possible third side could be.

COMPONENT SUMMARY

5.5.1 WARM-UP (~5 MIN.)

Would you rather? with given dimensions of pizza slices

5.5.3 GUIDED PRACTICE (10 MIN.)

Test the Triangle Inequality Theorem and given reasoning for answers

5.5.5 WRAP-UP (~5 MIN.)

~5 min.

5.5.2 COLLABORATION (15 MIN.)

Explore the Triangle Inequality Theorem with a partner

5.5.4 YOUR TURN! (10 MIN.)

Determine if given sides will form a triangle

RESOURCES

GLOSSARY

Key Terms:

- Triangle Inequality Theorem

Additional Terms:

- N/A

MATERIALS

- Dice (number cubes)
- (Optional) Paper
- (Optional) Ruler

ADDITIONAL PREPARATION

- N/A

TEACHER TIPS

COMMON MISCONCEPTIONS

- Students may think that when two sides equal the third, it will form a triangle.
- Students may think there is only one answer when determining a length for the third side of a triangle.

THINGS TO CONSIDER

- Make sure students are adding each pair of the shortest sides and comparing the sum to the third longest side.

LITERACY AND LANGUAGE ROUTINES

Poll the Class

Consider polling the class to see who has different answers for question 2d in component 5.5.2 Collaboration and use this information to group students. This will help quickly match students with different answers.

MATHEMATICAL THINKING AND REASONING (MTR) STANDARD

MA.K12.MTR.7.1 Real-World Contexts

Students are given problems that show the Triangle Inequality Theorem can be used in a real-world context. Encourage students to draw diagrams for these problems to help break them apart. This will help give context to the numbers students are given.

DIFFERENTIATE INSTRUCTION

This lesson provides a multitude of ways to differentiate. Notice that 5.5.2 Collaboration provides a kinesthetic entry point into the Triangle Inequality Theorem through having students utilize their arms to demonstrate triangle side lengths. This brief activity encourages students to engage with the material in an accessible way and provides students opportunity to physically see how the theorem is applicable in the real world, enhancing conceptual understanding as students work through the lesson.

5.5.1 WARM-UP: WOULD YOU RATHER?

DIFFERENTIATE INSTRUCTION

Consider having students sketch pizzas with the given dimensions to get a visual of each slice.



5.5.1 Warm-Up: Would You Rather?

The dimensions of two slices of pizza cut into triangles are given in the table.

	Slice A:	Slice B:
Height	2.8 inches (in.)	2.5 in.
Base	6.2 in.	6.6 in.

- Explain which slice of pizza you would rather eat.
Answers will vary. I want Slice B because the base is longer. I want Slice A because there is more area.
- Which slice of pizza has the larger area?
Slice A: $\frac{1}{2}(2.8)(6.2) = 8.68 \text{ in}^2$
Slice B: $\frac{1}{2}(2.5)(6.6) = 8.25 \text{ in}^2$
Slice A has the larger area.



5.5.2 Collaboration: Exploring the Triangle Inequality Theorem

- Put both of your elbows on your desk. Face your hands toward each other, then move them down until they are flat on the desk with only the fingertips touching. Slowly raise your hands up without moving your elbows.

Circulate to make sure students have palms down, arms in a line, middle fingertips touching.

- What do you notice about the length of your arms and the distance between your elbows?

Answers will vary. My forearms and hands add up to the length of the distance between my elbows.

- What do you wonder about creating a triangle with these lengths?

Answers will vary. I wonder why as I raise my hands I cannot make a triangle without moving my elbows closer to each other.

- Roll a die to pick 3 random numbers. Record your numbers in the blanks below. Then, circle true or false to indicate if the inequality is true or not.

Answers will vary.

a. $\frac{\underline{3}}{\text{Roll 1}} + \frac{\underline{4}}{\text{Roll 2}} > \frac{\underline{5}}{\text{Roll 3}}$ **true** or false

b. $\frac{\underline{4}}{\text{Roll 2}} + \frac{\underline{5}}{\text{Roll 3}} > \frac{\underline{3}}{\text{Roll 1}}$ **true** or false

c. $\frac{\underline{3}}{\text{Roll 1}} + \frac{\underline{5}}{\text{Roll 3}} > \frac{\underline{4}}{\text{Roll 2}}$ **true** or false

- Did you have any false answers?

Answers will vary.

- Find a partner who has a different answer for Part D. Discuss what you notice about how different values affected the truth of the inequalities. Summarize your discussion.

Answers will vary. When the inequalities are false, they will not form a triangle.

- Roll a die to pick 2 random numbers. Record your numbers on the blanks labeled for each roll. Leave any other lines blank.

Answers will vary.

a. $\frac{\underline{4}}{\text{Roll 1}} + \frac{\underline{6}}{\text{Roll 2}} > \underline{3}$

b. $\frac{\underline{6}}{\text{Roll 2}} + \frac{\underline{3}}{\text{Roll 1}} > \frac{\underline{4}}{\text{Roll 1}}$

c. $\underline{3} + \frac{\underline{4}}{\text{Roll 1}} > \frac{\underline{6}}{\text{Roll 2}}$

- Determine one value that will satisfy all three inequalities based on the values already inserted from rolling the die.

Answers will vary.

- Complete the statement.

The ◦ Triangle Inequality Theorem
◦ Pythagorean Theorem says any

◦ side
◦ angle of a triangle is ◦ more
◦ less than the

◦ difference
◦ sum of the other two sides.

5.5.2 COLLABORATION: EXPLORING THE TRIANGLE INEQUALITY THEOREM

TEACHER MOVES

This is to simulate attempting to form a triangle with sides (student's arms) that do not add up to be larger than the third side (the space between the elbows on the desk).

ENRICHMENT

- Help students make connections to triangles through questioning.
- What does it mean if the answers are equal to each other for question 2?
- What do you think it means for a triangle when the inequality is true?
- What do you think it means for a triangle when the inequality is false?

TEACHER MOVES

Poll the Class

Consider polling the class to see who has different answers for question 2d and use this information to group students. This will help quickly match students with different answers.

5.5.2 COLLABORATION: EXPLORING THE TRIANGLE INEQUALITY THEOREM (CONTINUED)

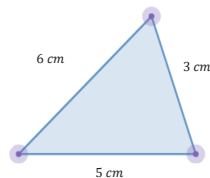
DIFFERENTIATE INSTRUCTION

Consider displaying this example and nonexample on the board to give students a large visual on what the side lengths of a triangle can be.



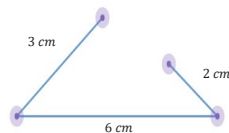
If any side is greater than or equal to the sum of the other 2 sides, a triangle will not be formed.

An example and a non-example are shown.



Forms a Triangle

$3 + 6 > 5$	✓
$3 + 5 > 6$	✓
$6 + 5 > 3$	✓



Does Not Form a Triangle

$2 + 3 > 6$	✗
$2 + 6 > 3$	✓
$3 + 6 > 2$	✓



5.5.3 Guided Practice: Applying the Triangle Inequality Theorem

1. Use the given sets of side lengths in the table to determine if a triangle can be formed. Use the explanation column to explain or show why or why not.

Side Lengths	Can the side lengths form a triangle?	Explanation
10, 2, 9	☐ Yes ☐ No	<i>The sum of any two sides is greater than the third.</i>
12, 2, 16	☐ Yes ☐ No	<i>12 + 2 is not greater than 16.</i>
6, 9, 12	☐ Yes ☐ No	<i>The sum of any two sides is greater than the third.</i>
7, 6, 13	☐ Yes ☐ No	<i>6 + 7 is not greater than 13.</i>

2. Two side lengths of a triangle are given. Determine a third side length that could be used to form a triangle.

Answers will vary.

a. 3, 4, 2

b. 1, 5, 5

c. 10, 20, 25

d. 300, 400, 350

- e. Find a partner who has different answers than you on any problem. Discuss whether both your answers are correct. Write a summary of your discussion.

Answers will vary. For problem 2a, my partner had 5 as the third side. Both of us are correct because both 2 and 5 could be used as the third side. $2 + 3 > 4$ and $3 + 4 > 5$.

5.5.3 GUIDED PRACTICE: APPLYING THE TRIANGLE INEQUALITY THEOREM

DIFFERENTIATE INSTRUCTION

For the explanation, consider allowing students to record themselves instead of writing for those who can better verbalize their answer.

TEACHER MOVES

Students have only seen whole-number side lengths. Consider challenging students to answer question 2 without using a whole number.

Encourage students to solidify understanding about the sides of triangles for question 2 through questioning. Why is there more than one answer for each set of side lengths in question 2? How can you determine the limits of the greatest side length possible for each set of side lengths in question 2?

5.5.4: YOUR TURN!

DIFFERENTIATE INSTRUCTION

If students are struggling with checking each pair of sides, have students highlight or draw lines to each measurement they are creating an inequality with. Students can also use the format with the three inequalities with blanks from the 5.5.2 Collaboration to help organize their thinking.

TEACHER MOVES

MA.K12.MTR.7.1: Mathematics in Real-World Contexts

Students are given problems that show the Triangle Inequality Theorem can be used in a real-world context. Encourage students to draw diagrams for these problems to help break them apart.

QUESTIONING

Have students create their own real-world triangle inequality problem and swap with a partner, then solve each other's problems.



5.5.4 Your Turn!

- Determine if the following sets of side lengths can create a triangle.
 - 3 in., 9 in., and 8 in.
Yes
 - 16 ft., 6 ft., and 2 ft.
No
 - 7 yd., 5 yd., and 10 yd.
Yes
- Adriana is creating a picture of a house with popsicle sticks. She wants to glue a roof in the shape of a triangle. Circle the set of sides that could form her triangular roof.

3 in., 5 in., 8 in.
4 in., 4 in., 9 in.
3 in., 3 in., 6 in.
5 in., 6 in., 7 in.
- Determine a third side that will form a triangle with the given side lengths.
8, 5, 4
Answers will vary.
- Maria is installing a garden fence in the shape of a triangle. She already has two pieces of fencing measuring 3 feet (ft.) and 4 ft. What is a possible length she can use for her third side?
Answers will vary. Any value greater than 1 and less than 7 is correct.

**5.5.5 Wrap-Up: Error Analysis**

Darion was asked to determine if the following side lengths would form a triangle: 4 ft., 6 ft., and 10 ft. Darion said that they would form a triangle because $4 + 6$ is not less than 10.

1. Help Darion understand and learn from his mistake. Write a brief note to Darion explaining what his mistake is and how he could learn from it.

Hey Darion, Although $4 + 6$ is not less than 10, it is not greater than 10 which it needs to be in order to form a triangle.

5.5.5 WRAP-UP: ERROR ANALYSIS**DIFFERENTIATE INSTRUCTION**

Consider having students act out this scenario with a partner. One person can be Darion explaining his idea at the board, while the other partner goes up after to show what needs to be corrected. This will allow an entry point for kinesthetic learners.